

Putnam POTD November 26, 2025

I. PROBLEM

For nonnegative integers n and k , define $Q(n, k)$ to be the coefficient of x^k in the expansion of $(1 + x + x^2 + x^3)^n$. Prove that

$$Q(n, k) = \sum_{j=0}^n \binom{n}{j} \binom{n}{k-2j}.$$

II. SOLUTION

We can factor $1 + x + x^2 + x^3 = \frac{x^4-1}{x-1} = (x+1)(x^2+1)$, so that the original expression is

$$P(x) = (x+1)^n (x^2+1)^n = \sum_{i=0}^n \binom{n}{i} x^i \sum_{j=0}^n \binom{n}{j} x^{2j}$$

then the coefficient for x^k is given by

$$Q(n, k) = \sum_{j=0}^n \binom{n}{j} \binom{n}{k-2j},$$

where the first term is the coefficient due to x^2 and the second due to x .